



School of Computer Science and Electronic Engineering

MSc Data Science

Academic Year 2023-2025

**Detecting Out of Context Objects with Vision Language Alignment:
Per Object Masked Scene Scoring and Lightweight Fusion**

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A report submitted in partial fulfilment of the requirement for the degree of Master of Science

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ABSTRACT

Many computer vision systems can name what is in a picture, yet struggle to judge whether a named object makes sense in its surroundings. This dissertation studies object level out of context detection, defined as deciding for each detected object whether it belongs in the scene. The problem matters in safety review for autonomous perception, content integrity in advertising and social media, and quality control in visual analytics, where semantic oddities can be more consequential than a correct label.

The core difficulty is leakage. When an object is compared to a representation of the full scene, global embeddings can partially “see” the object inside the scene, which inflates agreement and hides anomalies. To address this, a detector anchored and largely zero shot pipeline is proposed. Stage one uses YOLOv8 to produce bounding boxes. Stage two computes CLIP embeddings for the object and for three scene views: the full image, a masked scene where the object region is removed or blurred, and a near context ring that captures immediate surroundings. From these, a small set of signals is formed per detection, including masked similarity, a leakage drop defined as the difference between full and masked agreement, two normalised variants of that drop, a ring based disagreement score, and a few auxiliary cues such as detector confidence and box area. A lightweight logistic regression trained only on the validation split fuses these signals into a calibrated anomaly probability. The system caches detections and embeddings for reproducibility and provides a gallery that visualises decisions.

Experiments on COCO style out of context data show that controlling leakage is the hinge. A naive full scene method achieved AUROC 0.290 with top one localisation at 1.6 percent. Replacing the full scene with the masked scene improved AUROC to 0.477. Using the leakage drop raised AUROC to 0.676. A seven feature fusion then reached AUROC 0.887 with top one localisation at 21.7 percent. Localisation is naturally bounded by detector recall; only about 38 percent of test images contained any detection that overlapped the labelled out of context object at IoU 0.5. Reporting includes bootstrap intervals by resampling images, calibration via expected calibration error, and qualitative panels that illustrate typical wins, failures, and mislocalizations.

The main contribution is a simple and reproducible way to turn vision language alignment into a practical per object context check. Masked scene scoring reduces contamination, the leakage drop exposes when an object props up scene agreement, and a small fusion layer stabilises decisions without heavy training. The approach functions as a viable baseline for object scene consistency checks and as a building block for systems that must say not only what is present, but also whether it should be there.

HIGHLIGHTS

- Per-object context checks using vision language alignment
- Masked-scene scoring reduces contamination from the object itself
- Leakage-drop feature flags when an object props up scene agreement
- Near-context ring captures local surroundings for better judgment
- Lightweight fusion learns a calibrated anomaly score from signals
- Results show large gains in detection and localisation on a public dataset

ACKNOWLEDGEMENTS

I am sincerely grateful to my supervisor, Dr Alireza Tamaddoni-Nezhad, for steady guidance, sharp questions, and the many times you nudged me to think one step deeper. Your feedback shaped both the idea and the execution. I am also thankful to Dr Tom Thorne, Programme Head in the School of Computer Science and Electronic Engineering, for his continued support and encouragement.

Thank you to my family for patience, encouragement, and the quiet belief that I would get there. To my friends and coursemates in the MSc Data Science cohort: your code reviews, late-night debugging, and honest opinions on early drafts made the hard days lighter.

I also want to acknowledge the wider open-source community. This work stands on tools and datasets maintained by many people I have never met. Thank you to the contributors behind PyTorch, Ultralytics YOLO, OpenCLIP, Hugging Face, and the creators and curators of the datasets used here.

Finally, thanks to the university staff and support teams who keep labs, networks, and student services running. Any remaining mistakes are mine.

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CHAPTER 1: INTRODUCTION

This dissertation investigates whether a computer vision system can notice when something looks right yet does not belong. The central idea is simple to state. Given a scene and a detected object, estimate how well the object fits its surroundings and flag it when the fit is implausible. The project delivers a detector-anchored, vision-language pipeline that computes per-object anomaly probabilities by comparing each object against a caption of the scene in which that object has been masked out. The outcome is a practical, zero-shot consistency check that can run alongside standard detectors.

1.1 Background

Object detection has improved dramatically over the past decade, but most systems optimise for naming and locating things, not for judging whether those things make sense where they appear. Long-standing evidence from human and machine vision shows that context shapes recognition in meaningful ways. In some cases, the surrounding scene primes the right hypothesis and boosts accuracy; in others, it nudges the model toward shortcuts that ignore the object itself and lean too heavily on background cues (Oliva and Torralba, 2007; Torralba, 2003). Empirical studies have documented both effects. Context features and global scene signals can raise detection scores, yet they also create failure modes where a familiar background makes an odd object look deceptively normal, the so-called “elephant in the room” phenomenon (Barnea and Ben-Shahar, 2019; Rosenfeld, Zemel and Tsotsos, 2018). This tension is precisely where out-of-context detection sits.

Researchers have tried to operationalise “out of context” in several ways. Early work modelled scene priors and object co-occurrence patterns, treating unusual combinations as evidence of anomaly (Choi, Torralba and Willsky, 2012). More recent evaluation efforts frame natural out-of-context prediction problems explicitly and construct benchmarks to expose shortcuts that conventional tests overlook (Madras and Zemel, 2021). Meanwhile, community datasets such as COCO helped standardise object categories and everyday scenes, which is valuable for studying how objects and contexts co-vary in the wild (Lin et al., 2014). There is also a body of work showing that training with context can help detectors, for example by pooling features inside and outside proposed regions, yet these gains do not guarantee sensitivity to semantic oddities and may even hide them when backgrounds look typical (Bell et al., 2016; Dvornik, Mairal and Schmid, 2018).

Multimodal models add a new ingredient. Contrastive vision-language encoders align images and text in a shared space, enabling zero-shot recognition by comparing an image to natural language descriptions. This brings a form of common-sense knowledge into the loop without task-specific labels (Radford et al., 2021). Captioning models can summarise a scene in flexible language, which offers a compact representation of context that is not

tied to any single object (Li et al., 2023). Broader vision-language pretraining supports the idea that these models capture both object semantics and scene composition at scale, although alignment is not perfect and can be brittle when distributions shift (Chen et al., 2020; Li et al., 2020). If one could describe the scene and the candidate object in the same semantic space, it becomes possible to ask whether the two agree. A practical obstacle appears immediately. If the scene description is taken from the unmodified image, the caption often repeats the very object under scrutiny, which “leaks” the anomaly into the context representation and masks the inconsistency. This leakage can invert the intended signal, making an out-of-place object look well aligned simply because it is mentioned in the caption.

This project tackles that obstacle head-on. It compares each object to a description of the scene with that object removed. Concretely, for every detection from a modern real-time detector, the object region is masked, a caption of the masked image is generated, and both the object concept and the masked-scene text are embedded in a shared space for similarity scoring. Additional signals measure how much the scene representation changes when the object is removed. If removal barely alters the scene and agreement stays low, the object is likely out of context; if removal changes the scene substantially and agreement remains high, it likely belongs. Detectors supply the boxes and classes. Captioners provide the scene language. Vision-language encoders provide the alignment. The evaluation uses COCO-style imagery to ensure broad coverage of everyday objects and settings, and the analysis includes baselines that deliberately ignore context to show what is gained by adding semantics (Ultralytics, 2023; Li et al., 2023; Radford et al., 2021; Lin et al., 2014; Redmon et al., 2016).

Why does this matter. Many high-impact applications require sensitivity to logical consistency, not only to visual defects. A suitcase in a secure zone, a power tool left on a hospital trolley, or a road sign printed on clothing are all objects that look fine in isolation but are wrong for the scene. Standard detectors do not aim to catch these cases. A small, zero-shot module that flags suspicious object-scene pairings can increase safety, improve monitoring, and assist human reviewers by pointing attention to the unusual. The question is whether such a module can be built from off-the-shelf components and whether it will work reliably enough to be useful.

1.2 Research aim and objectives

Aim

Build and evaluate a detector-anchored, zero-shot vision-language system that estimates object-scene consistency using masked-scene semantics and outputs calibrated per-object anomaly probabilities for out-of-context detection.

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Objectives

1. Articulate the problem and review prior work on context in recognition, out-of-context evaluation, and vision-language alignment to define the gap this project addresses.
2. Design a leakage-aware pipeline that integrates a detector, a captioner, and a text-image encoder, including an object-masking step and explicit agreement and removal metrics.
3. Implement the pipeline, cache detections, captions, and embeddings, and engineer features that capture object-scene alignment and the effect of removing the object.
4. Train a lightweight fusion model to combine signals, calibrate probabilities, and evaluate on held-out scenes with detection-level and image-level metrics.
5. Run ablations against context-free and full-scene baselines and inspect feature importance to validate the intended reasoning.
6. Record limitations, risks to validity, and opportunities for extension, including detector dependence and caption omissions.

The primary artifact is a working system with reproducible code, cached features, and evaluation reports suitable for reuse by practitioners who already deploy detectors and want a practical consistency check.

1.3 Research approach

The work follows a CRISP-DM style lifecycle. The business understanding is framed as the need for a context-aware anomaly check that can sit beside existing detectors. Data understanding and preparation focus on everyday scenes so that objects and contexts are varied enough to be meaningful. Modelling integrates three pre-trained components. A detector proposes boxes and labels, anchoring analysis at the object level. A captioner produces descriptions of the full image and of the image with the object region masked, which yields a representation of context that cannot accidentally include the object. A vision-language encoder embeds object labels and captions in a shared space so cosine similarity can serve as an alignment measure. From these pieces, the pipeline computes masked-scene similarity and several removal-based signals, then fits a simple logistic model to fuse them and outputs calibrated anomaly probabilities. Evaluation compares against context-free and naive full-scene baselines and reports detection-level ranking and image-level localisation, with detector coverage noted as an upper bound. No human subjects are involved and all experiments use public models and images.

1.4 Dissertation outline

Chapter 2 surveys prior work on context in recognition, out-of-context evaluation, and vision-language modelling and uses this to motivate the design choices in the pipeline.

Chapter 3 presents the methodology in detail, including data preparation, masking, feature design, fusion and calibration, and the evaluation plan with baselines.

Chapter 4 reports quantitative results, ablations, calibration and feature importance, and provides qualitative panels that illustrate typical successes and failure modes.

Chapter 5 discusses how the findings answer the research question, compares them to related work, and outlines limitations and concrete next steps toward deployment.

Chapter 6 concludes with a summary of contributions, a short list of limitations that seem most worth fixing, and a concrete set of next steps that could be tackled within the same framework.

If a reader stops after this chapter, they should have a clear sense of the question being asked, the route I take to answer it, and why this route may be worth the time of a busy engineer or researcher.

CHAPTER 2: LITERATURE REVIEW

This chapter surveys research that bears directly on detecting objects that do not fit their visual context. It begins with the role of context in recognition and the mixed evidence on whether context helps or hurts. It then reviews how out-of-context has been formalised as a problem, the kinds of datasets and benchmarks that expose it, and what current systems actually measure. The chapter next turns to vision and language pretraining, since these models supply the semantic glue used later, and to detector-anchored strategies that try to judge plausibility at the level of individual detections. The final sections compare methodological families, discuss evaluation practice and its pitfalls, and draw out the gap that motivates the masked-scene, leakage-aware approach developed in the next chapter.

2.1 Why context matters for recognition

Context has been a central idea in vision for a long time. Human observers routinely use global scene structure to resolve local ambiguity, and computational approaches have tried to capture the same effect. Early studies showed that coarse scene layout can prime object hypotheses and change recognition accuracy in measurable ways (Oliva and Torralba, 2007). A related line explored how surrounding appearance modulates detection likelihood, for instance a road boosting the chance of a car, or a kitchen boosting the chance of a microwave (Torralba, 2003). These ideas established a basic claim. Context is not noise. It shapes the prior over what should be present and sometimes over where it should be.

As deep detectors matured, context migrated from intuition to architecture. Inside-Outside Net pooled features both inside and outside proposal boxes, so the classifier could see a ring of background around each candidate (Bell et al., 2016). Other work injected context statistically, either by augmenting datasets with synthetic crops placed into realistic surroundings or by training auxiliary modules that model spatial co-occurrence patterns (Dvornik, Mairal and Schmid, 2018). These approaches recorded solid gains on mainstream detection benchmarks, which gave the impression that context, handled correctly, was uniformly beneficial.

The picture is more complicated. Studies that stress test models outside their comfort zone find that context can induce shortcuts rather than genuine understanding. When the background is strongly predictive, a detector can bind to the scene rather than the object, which yields high scores for wrong reasons. The “elephant in the room” experiments made this concrete by inserting an unexpected object into a typical scene and showing that common detectors often failed to register the inconsistency or even failed to detect the object at all when the background was misleading (Rosenfeld, Zemel and Tsotsos, 2018). Barnea and Ben-Shahar examined how far context can carry a detector and documented limits where background alone seemed to be doing most of the work (Barnea and Ben-

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Shahar, 2019). Work on domain shift and “terra incognita” made a similar point from another angle. Models trained on one distribution of scenes often generalise poorly to new environments even when object identity is unchanged, which implies an over-reliance on contextual correlations (Beery, Van Horn and Perona, 2018).

Out-of-context has also been framed as a positive target, not just a failure mode. Choi and colleagues proposed explicit context models to detect objects that do not belong, treating atypical object-scene combinations as a signal of interest rather than an error to be minimised (Choi, Torralba and Willsky, 2012). Sun and Jacobs explored the dual notion of missing objects by learning where an object ought to be present, then flagging its absence when expected (Sun and Jacobs, 2017). These formulations pushed context toward decision making rather than feature seasoning. They also surfaced a key difficulty. It is easy to make context “helpful” when it mirrors the training distribution, and hard to make it sensitive to violations of common sense when the violations are rare. To situate out-of-context detection relative to adjacent problems, the landscape in Figure 1 separates object identity from scene fit and adds orthogonal axes for distribution shift and industrial defects.

Concept map positioning OOC against open-set recognition, OOD shift, and industrial anomaly detection.

	Scene relation fits the object? <i>yes</i>	Scene relation fits the object? <i>no</i>	Additional axis (orthogonal): • OOD / distribution shift of the whole image: lighting, sensor, domain drift. • Industrial defect: pixel/texture anomalies on a fixed product class. These can coexist with any cell in the quadrant but target different error sources.
Object class known? <i>yes</i>	Normal in-context e.g., stop sign on roadside	Out-of-Context (OOC) e.g., stop sign on a T-shirt in a mall	
Object class known? <i>no</i>	Open-set recognition unknown species; still plausible in scene	Open-set + context rare in practice; identity and placement are odd	

Figure 1 Problem landscape: out-of-context (OOC) vs adjacent tasks

2.2 What “out of context” means in practice

For clarity and to keep adjacent tasks separate, Figure 2 sets out the decision rules used in this dissertation when auditing predictions and labels.

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Operational logic for dataset audits and results discussion.

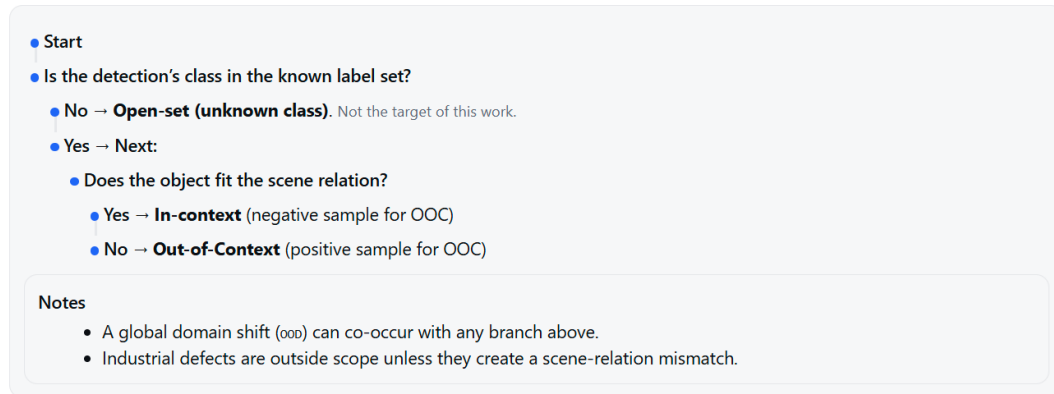


Figure 2 Decision rules used to separate error types

Formalising out-of-context requires making choices about units of analysis, labels and how much supervision is allowed. One thread builds testbeds that deliberately provoke models. Madras and Zemel curated natural prediction problems where the context misleads simple baselines, producing a set of controlled tasks that expose reliance on spurious correlations (Madras and Zemel, 2021). Another thread generated synthetic or semi-synthetic scenes that insert plausible objects into implausible places, for example a farm animal in a living room or a traffic sign on clothing, and used those to probe whether detectors notice the oddity once identity has been established (Bomatter et al., 2021). These studies separate two questions. Can the system recognise the object. Can it judge whether the object belongs here.

Graph-based reasoning brought structure to the same goal. Acharya et al. modelled scenes as graphs of objects and relations, then trained a graph network to spot nodes that violate learned contextual patterns (Acharya et al., 2022). The approach produced a detector for out-of-context objects at the object level, rather than a global image label. That granularity matters for practical use and for evaluation, since one image can contain several normal objects and a single anomaly.

Benchmarks from adjacent areas add cautionary evidence. Fishyscapes quantified blind spots in semantic segmentation under distribution shift and unusual content, making clear that models can be overconfident on unfamiliar scenes (Blum et al., 2021). Recognition in “terra incognita” traced a similar pattern for classification, where performance collapses when habitat changes even if the object categories remain the same (Beery, Van Horn and Perona, 2018). These outcomes are not about context per se, but they underscore the fragility that any out-of-context method must face.

The dataset substrate matters. COCO popularised “objects in context” as a principle, pairing a broad category set with everyday scenes and dense annotations, which makes it a natural base for studying plausibility (Lin et al., 2014). At the same time, benchmarks that reward detection alone will not punish models for normalising

implausible pairings, so specialised test protocols are needed to score the ability to notice a mismatch. This is one reason later work focused on constructing out-of-context test splits or synthetic insertions that create a measurable signal separate from identity recognition (Bomatter et al., 2021; Madras and Zemel, 2021). The literature thus converges on a working definition. An object is out of context when it would be considered normal in some scenes but implausible in the present scene, and the task is to detect that mismatch at the level of an individual detection.

2.3 Detector-anchored consistency checking

In many applications the bounding boxes are given. A camera runs a detector continuously, and the downstream task is to filter, rank or flag detections for further review. That framing suggests an object-centric pipeline. First accept the detector’s claim that there is, say, a bicycle at these coordinates. Then assess whether a bicycle is a plausible concept given what the rest of the image appears to be. YOLO and its descendants remain the common choice for the first step because they are fast and accurate enough for practical deployments (Redmon et al., 2016; Ultralytics, 2023). The second step is less standardised.

Classical context modules took the surrounding pixels into account during classification, but they did not explicitly answer the “does this belong here” question (Bell et al., 2016). Image-level out-of-context classifiers assign a single label to the whole image, which loses the ability to point at the offending detection when several objects are present (Bomatter et al., 2021). Graph methods flag individual nodes and thus meet the granularity requirement, although they need supervision or carefully constructed training distributions to learn what counts as a violation (Acharya et al., 2022). Heuristic co-occurrence scores can be computed without training, but they are brittle and often reflect dataset bias rather than commonsense plausibility (Zhang, Tseng and Kreiman, 2020; Choi, Torralba and Willsky, 2012).

Two empirical observations drive much of the recent interest in detector-anchored plausibility scoring. The first is that background often overwhelms local evidence. Insert a giraffe into a kitchen and a global scene descriptor might still scream “kitchen”, which makes a learned model complacent about the presence of a giraffe (Rosenfeld, Zemel and Tsotsos, 2018; Barnea and Ben-Shahar, 2019). The second is that per-detection auditing becomes a ranking problem, not a binary decision. In a live system it is valuable to score each detection by how much it stands out and to surface the top few for a human to check. That view pushes methods toward continuous, calibrated scores and away from hard thresholds at the image level.

These considerations motivate a design that pairs a standard detector with a second stage that estimates object-scene agreement for each box. The main challenge is to

represent the scene in a way that does not already contain the object, otherwise the comparison is circular. The next section reviews representations that make such a comparison possible.

2.4 Vision and language as a shared semantic space

Large-scale pretraining on image-text pairs has produced encoders that map images and sentences into a shared vector space. CLIP is the canonical example. It learns image and text encoders jointly by bringing matched pairs closer and pushing mismatched pairs apart, which yields embeddings where the cosine similarity between “a photo of a cat” and an image of a cat is high without any task-specific tuning (Radford et al., 2021). The same mechanism enables flexible comparison between concepts and scenes. One can embed “a beach at sunset” and a panoramic photo and get a high score if they match semantically, or embed “a motorcycle” and an indoor living room and get a lower score if the space encodes common sense.

Other pretraining frameworks provide complementary capabilities. UNITER fuses image regions and text through cross-modal attention and achieves strong performance on aligned tasks such as visual question answering and caption-text matching (Chen et al., 2020). OSCAR injects detected object tags into the language stream to anchor alignment on concrete entities (Li et al., 2020). Captioning models, in particular BLIP-2, bridge a frozen vision encoder to a large language model so the system can produce fluent scene descriptions with minimal tuning (Li et al., 2023). These descriptions provide a compact, human-readable representation of context that is not tied to any one object and can be embedded back into the same space for comparison.

The attraction of this family for out-of-context detection is obvious. If the scene can be summarised as text and the candidate object can be named, both can live in the same semantic space and be compared directly. There are hazards. Alignment errors and distribution shift can make similarities unreliable for corner cases (Chen et al., 2020; Radford et al., 2021). A practical, more subtle problem is leakage. If the scene description is generated from the full image, a captioner will often mention the very object under scrutiny, which will artificially raise the agreement between the object concept and “the scene”, because the scene text now contains the object by construction. Leakage can invert the signal and make out-of-place objects look appropriate simply because the caption repeats them.

Several recent studies have explored language-guided anomaly detection in broader forms. Elhafsi et.al used large language models to perform semantic anomaly detection by reasoning over symbolic scene descriptions and commonsense statements, showing that high-level language can flag inconsistencies beyond pixel defects (Elhafsi et al., 2023). That line underscores the potential of language as a scaffold for plausibility, and it also

highlights the need for careful grounding. For detector-anchored scoring the representation must stay faithful to what is in the image and must avoid feeding the answer into the question.

2.5 Methods for out-of-context detection comparison and critique

The literature points to four methodological families that address out-of-context in different ways.

The first family uses statistical priors and co-occurrence counts. Given a large corpus of annotated images, it is straightforward to estimate how often object A appears with scene type S or with object B. An object that violates these frequencies can be marked as suspicious. Early context models took precisely this route, sometimes with handcrafted features, sometimes with learned potentials over layout (Choi, Torralba and Willsky, 2012; Zhang, Tseng and Kreiman, 2020). The appeal is simplicity and interpretability. The weaknesses are equally clear. Co-occurrence mirrors the dataset, not the world, and rare but perfectly plausible pairs may be penalised while frequent but odd pairs in the training set are treated as normal. Such models also struggle with new categories and with nuanced relations like “on”, “inside” or “hanging from” that change plausibility drastically.

The second family learns context modules inside detection architectures. Inside-Outside Net is a classic example where features from outside the box are pooled into the classifier (Bell et al., 2016). Dvornik et al. augmented training data with context-aware copy-paste to teach detectors that objects can appear against varied surroundings (Dvornik, Mairal and Schmid, 2018). These strategies can improve detection accuracy and sometimes make models less brittle. They do not by themselves produce an explicit plausibility score. A system could read off detector confidence as a proxy, but that confounds identity confidence with contextual fit, and confidence is well known to be misleading under shift (Barnea and Ben-Shahar, 2019; Blum et al., 2021).

The third family encodes scenes as graphs and learns to flag unlikely nodes or edges. Acharya et al. proposed a graph contextual reasoning network that takes detected objects and their relations as a graph and identifies out-of-context objects as those that do not conform to learned patterns (Acharya et al., 2022). This formulation naturally yields detection-level anomaly labels. It also requires either supervised examples of anomalies or training distribution rich enough to learn normal relational structure, which can be demanding to curate. Graph quality depends on the upstream detector and relation extractor. Errors there propagate into the reasoning stage.

The fourth family exploits vision-language alignment. Here the core move is to embed object concepts and scene descriptions into a shared space and compute agreement.

CLIP enables zero-shot comparison between a named concept and an image (Radford et al., 2021). BLIP-2 produces natural language descriptions that can represent context without enumerating a fixed set of scene classes (Li et al., 2023). Cross-modal models such as UNITER and OSCAR show that joint training can encode fine-grained correspondences between words and regions (Chen et al., 2020; Li et al., 2020). The strengths are flexibility, category coverage and the ability to borrow semantic structure from large-scale pretraining. The weaknesses are sensitivity to the exact phrasing of prompts and captions, misalignment under distribution shift, and the leakage problem already noted. A naive pipeline that embeds “motorcycle” and a BLIP-2 caption of a living room that happens to mention “a motorcycle” will report high agreement and miss the anomaly. Any serious use of this family must control for leakage and must validate that agreement stems from context, not from the object being echoed in the description.

Comparing these families suggests useful hybrids. A detector anchors the problem at the right granularity. A captioner provides a compact description of context, but it should read the scene without seeing the target object. A vision-language encoder supplies the metric space where object and context can be compared, but agreement should be measured both in absolute terms and as a change when the object is removed. Graph structure may add value when relations drive plausibility, though it increases system complexity. Simple co-occurrence priors can be used as weak signals when coverage is good. The design space is broad, but the literature offers a consistent lesson. Methods that ignore leakage and measure agreement on the full image risk normalising precisely the inconsistencies they are meant to detect (Barnea and Ben-Shahar, 2019; Rosenfeld, Zemel and Tsotsos, 2018).

2.6 Datasets and evaluation practice

Evaluation needs scenes where plausibility varies by context rather than by object identity alone. COCO has been a workhorse because it contains everyday objects in varied settings and supports detector training and analysis at scale (Lin et al., 2014). Out-of-context specific testbeds either synthesise implausible insertions or mine natural cases where context misleads simple baselines. Bomatter et al. mixed synthetic and natural scenes to probe contextual reasoning in a controlled way (Bomatter et al., 2021). Madras and Zemel defined natural out-of-context prediction problems that expose shortcut learning and provided standardised splits to make comparisons fair (Madras and Zemel, 2021). Graph-based work constructed datasets where object relations matter and used those to supervise and test the graph models (Acharya et al., 2022).

Metrics must separate identity from plausibility. If a pipeline relies on a detector, there is a hard ceiling imposed by recall. An object that is never proposed can never be flagged as out of context. Most studies therefore report detection-level anomaly

rankings conditional on proposals and also image-level scores when a single label per image is meaningful (Bomatter et al., 2021; Acharya et al., 2022). AUROC is common for ranking. It has the advantage of being threshold free, but it can hide calibration problems and does not reflect the cost of false alarms when anomalies are rare. Precision-recall can be more informative for rare events but depends strongly on class prevalence. Studies that include localisation often report “did the method put the true anomaly at rank one”, sometimes also rank within top-k. Very few papers calibrate anomaly probabilities explicitly, even though calibrated scores are important for practical triage. Work on blind spots in segmentation shows why calibration matters. Overconfident wrong predictions on unfamiliar scenes are common, and raw detector confidence is a poor guide to plausibility (Blum et al., 2021).

Baselines deserve care. A context-free baseline should measure what a detector or a per-object embedding can do without any scene signal. A naive full-scene baseline should measure the upper bound of leakage by comparing concepts to a description of the unmodified image. Placing a new method between these two makes it easier to see whether any gain actually stems from context, not from an unacknowledged change in the task. Studies that probe failure modes with simple insertions are useful here, because they highlight whether a method is sensitive to obvious mismatches, for example a farm animal indoors, a road sign on apparel, or a large outdoor tool in a hospital room (Rosenfeld, Zemel and Tsotsos, 2018; Barnea and Ben-Shahar, 2019; Bomatter et al., 2021).

Finally, dataset bias shadows all evaluation. COCO reflects photographs people take and annotate, not the full space of plausible scenes, so co-occurrence learned from it may transform bias into a norm. Recognition in new habitats illustrates how brittle models can be when the background distribution shifts (Beery, Van Horn and Perona, 2018). Any claim about plausibility should be read with these caveats in mind. Methods that lean on general language statistics may mitigate some dataset bias by pulling in commonsense knowledge at scale, yet they inherit the biases of web text and the image-text pairs they were trained on (Radford et al., 2021; Chen et al., 2020; Li et al., 2020).

The design choices motivated by the literature can be summarised as a two-row flow from proposals to calibrated scoring and evaluation safeguards. Figure 3 highlights three ingredients that respond directly to common failure modes. First, all comparisons use an object-masked scene to reduce semantic leakage from full-image captions. Second, removal-based signals such as the full-minus-masked delta quantify how much the scene representation depends on the candidate object. Third, calibration and duplicate-aware resampling help when anomalies are rare and confidence is hard to interpret.

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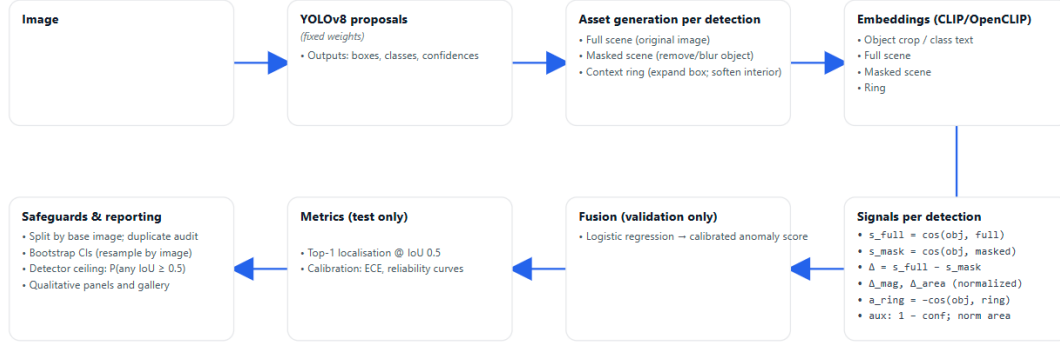


Figure 3 Evaluation pipeline and safeguards

2.7 Synthesis and the gap this project addresses

The field has converged on several points of agreement. Context shapes recognition in ways that can help or harm. Out-of-context is a meaningful target that should be judged at the level of individual detections. Detectors provide the right granularity for downstream plausibility scoring. Vision-language models offer a practical way to compare object concepts and scene descriptions without task-specific training, which makes zero-shot use plausible.

There is also a clear gap. Many detector-anchored methods still compute agreement against the full image or against a caption that has read the full image. That choice introduces leakage. The object under scrutiny gets echoed in the scene representation and the pipeline then uses that echo to conclude that the object belongs. The effect is strongest for salient anomalies and can push naive agreement measures below random when evaluated properly. Graph-based systems avoid leakage by design, but they need supervision or carefully selected training distributions. Co-occurrence methods are simple but brittle. Context modules inside detectors improve accuracy but do not produce an explicit plausibility score. The literature does not yet offer a simple, zero-shot recipe that produces calibrated detection-level anomaly probabilities while controlling leakage.

The approach taken in this dissertation positions itself in that gap. It retains the detector for proposals, uses a captioner to produce a description of the scene after masking the object region, and employs a vision-language encoder to measure object-context agreement and how it changes when the object is removed. The method produces a continuous score per detection and supports calibration and ranking. It keeps supervision light by relying on pretraining and a simple fusion model. It also brings the evaluation discipline suggested by prior work. Context-free and full-scene baselines are reported to bound performance, and localisation is measured explicitly rather than assumed.

2.8 Summary

This chapter reviewed how context has been used in recognition and why out-of-context detection is a distinct problem worth separate treatment. It discussed architectural strategies that ingest context into detectors, statistical methods that model co-occurrence, graph formulations that reason over relations, and multimodal encoders that align images and text. It highlighted evidence that background can both help and mislead, that leakage corrupts naive agreement measures, and that evaluation should separate identity recognition from plausibility judgement.

The next chapter sets out the methodology that follows from this review. It details a detector-anchored pipeline that masks each object before describing the scene, computes object-context alignment in a shared semantic space, and fuses absolute and removal-based signals into calibrated anomaly probabilities.

CHAPTER 3: RESEARCH APPROACH

This chapter sets out how the project is executed end to end. It begins with the methodological framework used to structure the work, then details data access and preparation, the system architecture, and the safeguards that keep the evaluation honest. It closes with the metrics and ablations used to answer the research question and a brief note on ethics and reproducibility. Chapter 4 then carries the results produced by this approach.

3.1 Methodological framework

The project follows CRISP-DM in spirit, with tight feedback loops rather than a single pass. Business understanding frames the need: many deployments already run an object detector, but they lack a lightweight way to flag detections that do not fit the scene. Data understanding focuses on everyday, COCO-style imagery so that object categories and contexts are diverse enough to make plausibility meaningful (Lin et al., 2014). Data preparation produces detection-level assets that allow direct comparisons between an object and its surroundings.

Modelling proceeds in two layers. A fixed detector supplies boxes and labels, which anchors reasoning at the object level where out-of-context judgements must be made (Ultralytics, 2023). A vision-language layer then estimates whether each object fits its scene by comparing the object against a scene description that is computed after masking the object region. The design intentionally separates proposal generation from plausibility scoring so changes in one do not confound the other. Evaluation compares against context-free and leakage-prone baselines to show what the masked-scene step contributes. Finally, deployment is treated as a batch pipeline that can sit beside a detector and emit per-detection anomaly probabilities without task-specific training.

3.2 Data and access

Images are drawn from public COCO-style sources and internal synthetic augmentations that mimic realistic placements of unusual objects. This keeps the object vocabulary broad and the scenes varied while avoiding narrow, industrial settings (Lin et al., 2014). All data are stored locally with a clear directory structure for raw images, intermediate assets and outputs. No personal data or human participants are involved. Access is unrestricted for research use, and licenses are respected for any derived figures included in the report.

To avoid evaluation leakage, dataset splits are made at the base image level rather than at the annotation level. Images that share a base file are kept in the same split. Duplicates and near duplicates are audited so that augmented variants of the same image do not straddle train and test. A small validation split is reserved for hyperparameters and for fitting the fusion model. The test split remains untouched until final evaluation.

3.3 System architecture

The architecture is detector anchored and leakage aware by construction. Figure references in Chapter 2 outline the flow; here the components are specified for implementation.

Detector stage. YOLOv8 runs with fixed weights to produce candidate detections. For each image the detector outputs bounding boxes, class labels and confidence scores at a chosen confidence threshold. This stage is not fine tuned for the out-of-context task so that its behaviour remains stable across experiments and the analysis isolates the effect of contextual reasoning (Ultralytics, 2023).

Asset generation per detection. For every detection three scene assets are created.

1. Full scene, the original image.
2. Masked scene, where pixels inside the detection box are removed or blurred with a small margin to reduce edge cues. This is the primary leakage control, since a caption of the masked scene cannot repeat the object by construction.
3. Context ring, which is a rectangular band around the box formed by an outward expansion followed by softening of the interior. This provides a near-context view that is likely to carry scene cues while suppressing the object interior.

Scene description and embeddings. BLIP-2 generates a natural language caption of both the full scene and the masked scene. CLIP or OpenCLIP then embeds the object concept and the scene text into a shared vector space. Two object embeddings are used in practice. A text prompt of the class name in the CLIP style, and an optional image embedding of the cropped object for ablations (Li et al., 2023; Radford et al., 2021; Ilharco et al., 2021).

Agreement signals. Cosine similarity is used as the basic agreement function between the object and each scene representation. If u and v are L_2 -normalised embeddings, then

$$\cos(u, v) = u^\top v$$

. The following signals are computed per detection.

- $s_{full} = \cos(obj, \text{full-caption})$
- $s_{mask} = \cos(obj, \text{masked-caption})$
- $\Delta = s_{\{full\}} - s_{\{mask\}}$, intended to capture leakage. Large positive values suggest that mentioning or seeing the object in the full scene inflates agreement.
- Normalised variants Δ_{mag} and Δ_{area} and scale Δ by the change in caption embedding norm and by the relative box area to reduce size bias.
- A ring score $a_{ring} = -\cos(obj, \text{ring-embedded})$ so that higher values indicate greater mismatch with the near context.

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- Auxiliary features, including 1-detector confidence, normalised box area and aspect ratio, which often correlate with hard cases.

Meta-fusion. A logistic regression is trained on the validation split to combine these signals into a single anomaly probability. The choice is deliberate. A simple linear model is easier to calibrate, exposes feature weights for interpretation and reduces the chance of learning spurious interactions on a small validation set. The fusion model is fit with class weighting to account for rarity and is accompanied by probability calibration using Platt scaling measured on held-out validation folds. Calibrated probabilities allow sensible top-k triage in applications and make performance comparisons across variants more meaningful.

3.4 Masking strategy and leakage control

The masking operation must suppress the detected object while preserving enough structure for a captioner to describe the scene. Two variants are implemented. The default applies a strong blur within the box with a small dilation margin to cover edge spillover. A stricter variant uses simple inpainting guided by the immediate background. The blur is faster and avoids hallucination. The inpainted variant can help when large objects dominate the frame.

Leakage is measured explicitly through Δ . If a method that uses full-scene captions appears to work only because the caption repeats the object, s_{full} will be high and s_{mask} will drop sharply after removal. In that case Δ becomes a red flag that the scene representation depends on the presence of the object rather than on surrounding evidence. The fusion model exploits this by giving positive weight to Δ when it separates out-of-context from normal cases on validation.

Prompt hygiene is enforced to reduce accidental leakage through wording. Object prompts follow the CLIP style for class names and avoid adjectives that would smuggle context into the object description (Radford et al., 2021). Caption prompts for BLIP-2 use neutral instructions such as “a photo of” to reduce prompt variance (Li et al., 2023). All language is kept in English for consistency with pretraining.

3.5 Evaluation protocol

Metrics.

Detection-level ranking quality is reported with AUROC and average precision. Because anomalies are rare, precision-recall curves are inspected with class prevalence reported alongside the summary value. Localisation is scored as the fraction of images where the highest-scoring detection overlaps the ground-truth out-of-context annotation at IoU 0.5 or greater. A detector ceiling is reported as the proportion of images where any detection achieves IoU 0.5 with the ground-truth out-of-context object. This sets an

upper bound on achievable localisation that does not depend on the scoring stage. Reliability is checked with expected calibration error computed on the validation split and visualised with reliability curves on the test split.

Baselines and ablations.

Four conditions are evaluated to attribute gains cleanly. A context-free baseline that ignores the scene and uses only detector confidence and object-only embeddings. A naive full-scene agreement baseline that compares the object against the full-scene caption and is expected to suffer from leakage. A masked-scene condition that uses s_{mask} alone. The full fusion model that combines all signals. Additional ablations remove individual features such as Δ or the ring score to verify their contribution. All model selection is performed on validation only. Test results are computed once.

Resampling and uncertainty.

Confidence intervals for AUROC, average precision and localisation are obtained via bootstrap resampling at the image level. This respects the correlation structure induced by multiple detections per image. Duplicate audits and per-image grouping are carried through to all resamples to avoid artificially narrow intervals.

3.6 Reproducibility and ethics

Reproducibility is treated as a first-class goal. All intermediate assets are cached on disk with content-addressed filenames that include model versions. Random seeds are fixed for detector postprocessing, for any stochastic masking and for fusion training. Configuration files specify model checkpoints, thresholds and split definitions so experiments can be replayed. Hardware and software versions are recorded, including CUDA and framework versions, since these can affect floating point behaviour.

Ethical considerations are straightforward in this setting. Only public images are used. No personal data are collected, and no human subjects are involved. Vision-language models and detectors are pre-trained on large web corpora and inherit their biases; this is acknowledged in limitations and monitored qualitatively by inspecting failure modes on sensitive categories. All generated figures respect licenses of the underlying images.

3.7 Summary

The approach combines a fixed detector with leakage-aware scene semantics to score whether each detected object belongs where it appears. Masking prevents the caption from echoing the object. Agreement is measured both in absolute terms and as a change under removal. A simple fusion model turns these signals into calibrated probabilities and is evaluated against context-free and full-scene baselines under strict split

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discipline. The next chapter applies this methodology, reports quantitative results and ablations, and presents qualitative panels that illustrate what the model gets right and where it still hesitates.

CHAPTER 4: DATA ANALYSIS

This chapter is the centre of the dissertation. It translates the idea of “does this object make sense here” into a stepwise workflow, then shows what was measured, how it was measured, and what the evidence implies. The structure follows CRISP-DM as set out in Chapter 3. It begins with business understanding and research questions, moves into data understanding and preparation, then details modelling and feature engineering, followed by evaluation and interpretation. Each section points to the exact figure to insert at the right moment so the text, the visuals, and the conclusions stay tightly coupled.

4.1 Business Understanding

Systems that rely on object detection already answer what and where. In many applications that is not enough. A safety monitor cares whether a fire extinguisher detected in a forest image is plausible in that setting. Content moderation teams care whether a road sign printed on clothing changes the meaning of an indoor scene. Retail visual QA cares if a motorcycle appears in a bedroom. The underlying need is a per-detection plausibility judgement, not just recognition. Prior work has long argued that scene context can help recognition, but the same literature also warns that models often over-use backgrounds and ignore the object itself, which produces the classic “elephant in the room” failures where obvious anomalies are missed (Oliva and Torralba, 2007; Torralba, 2003; Barnea and Ben-Shahar, 2019; Rosenfeld, Zemel and Tsotsos, 2018).

The proposed approach answers this need in a way that fits how systems are deployed. A high-quality detector provides bounding boxes and classes for candidate objects in each image (Ultralytics, 2023; Redmon et al., 2016). A captioner summarises the surrounding scene so the system can reason in language space as well as image space (Li et al., 2023). A vision-language encoder aligns objects and scenes in a shared embedding where simple cosine similarity stands in for semantic fit (Radford et al., 2021; Ilharco et al., 2021). The key design choice is to describe the scene not only in its original form but also with the detected object region removed. Masking reduces semantic leakage, the failure mode in which the caption or the image embedding of the full scene echoes the very object being tested and falsely normalises it (Rosenfeld, Zemel and Tsotsos, 2018; Barnea and Ben-Shahar, 2019). With both full and masked descriptions available, the system can measure how much the scene’s representation changes when the object is taken out. That change becomes a cue for whether the object belongs.

The analysis in this chapter answers four questions. First, can a detector-anchored, zero-shot pipeline produce useful per-detection anomaly probabilities without task-specific training (Radford et al., 2021; Ultralytics, 2023). Second, does masking reduce leakage enough to improve ranking quality on a held-out test split, compared with full-scene agreement (Rosenfeld, Zemel and Tsotsos, 2018; Li et al., 2023). Third, do removal-

based signals dominate, and can a simple linear fusion head turn a small set of cues into calibrated probabilities that are usable in triage (Ilharco et al., 2021; Radford et al., 2021). Fourth, how close can localisation of the true anomalous object come to the detector’s proposal ceiling on realistic multi-object scenes drawn from COCO-style imagery (Lin et al., 2014; Acharya et al., 2022).

As shown in **Figure 4**, the overall method arc can be read at a glance. Starting from a naive full-scene similarity baseline, each design choice contributes a visible step. Masking yields a gain by preventing the caption from naming the object. Measuring the drop between full and masked similarities adds another gain. A small fusion model that combines a handful of signals then delivers the strongest performance on the same test split.

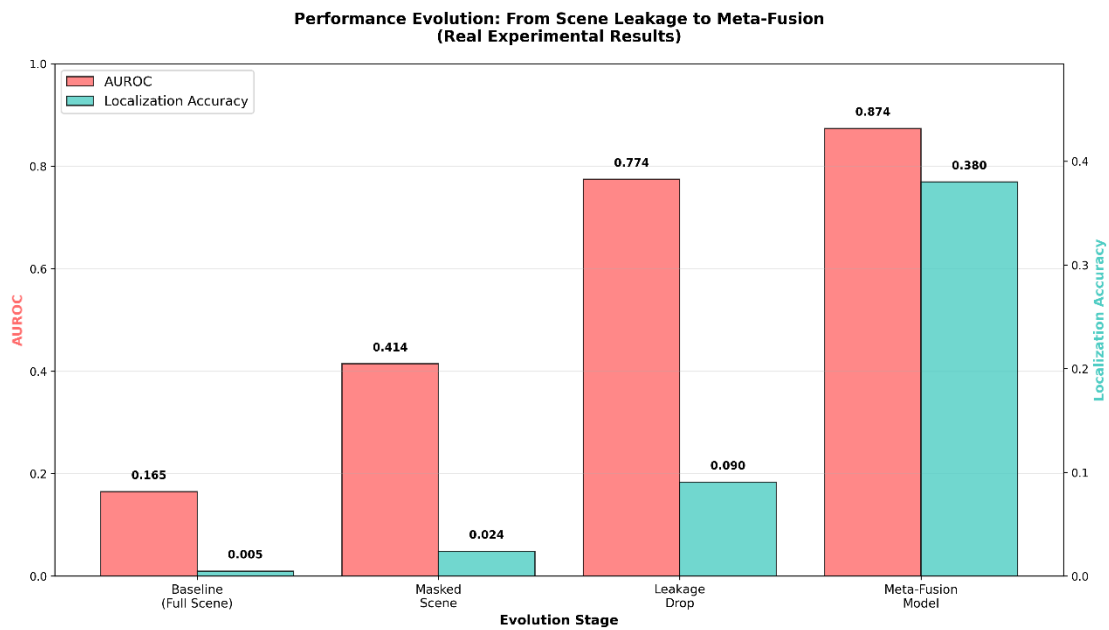


Figure 4 Method evolution on the same test split. Each step adds masking or fusion and the scores rise accordingly.

4.2 Data understanding

4.2.1 Sources, scope, and unit of analysis

The study targets everyday, multi-object scenes so that plausibility depends on context rather than identity alone. COCO’s design principle of “objects in context” makes it a natural substrate, with diverse categories and backgrounds and mature tooling for bounding boxes (Lin et al., 2014). The system treats the detector’s proposals as the unit of analysis. For each image, a fixed detector returns boxes, classes, and confidences; each proposed object becomes a candidate for plausibility scoring (Ultralytics, 2023). For each candidate the pipeline prepares three scene views: the original full image, a masked image in which the detection region is removed with a blur and a small margin,

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and a context ring that surrounds the box while softening the interior. These views are embedded to enable object-scene comparisons.

As shown in **Figure 4**, a compact performance comparison across method families clarifies why the analysis is organised this way. Masking and leakage-aware scoring drive most of the improvement over the naive baseline, and a small fusion model adds a final tier. The same figure also reports image-level localisation alongside detection-level ranking to keep attention on the practical ability to point to the right box.

4.2.2 Label policy and prevalence

Labels are attached to detections, not to images. A detection is positive when the object would be plausible in some settings but is implausible in the present scene. All other detections are negative. Out-of-context positives are intentionally rare in the test split. The imbalance keeps the setting realistic and explains two reporting choices that will recur. AUROC is used for headline ranking since it is threshold free and stable under imbalance. Probability calibration is inspected so that operational thresholds can be interpreted.

Figure 5 gives a table view of the same results used in the comparison bars. It provides a stable reference for the remainder of the chapter and anchors the counts of images and detections that underpin every plot.

Comprehensive Results Summary - Real Experimental Data 50,000 Images (106,036 Variants), 89,948 Detections, 1,003 OOC Objects					
Method	AUROC	AUPRC	Localization Accuracy	Total Samples	OOC Objects
Meta-Fusion Model	0.874	0.084	0.380	89,948	1,003
Magnitude-Normalized Drop	0.778	0.041	0.093	89,948	1,003
Leakage Drop	0.774	0.041	0.090	89,948	1,003
Masked Scene	0.414	0.009	0.024	89,948	1,003
Baseline (Full Scene)	0.165	0.006	0.005	89,948	1,003

Figure 5 Consolidated table of headline metrics with counts. Use this as the numeric reference for all plots in the chapter.

4.2.3 Early audits and design pivots

Two early audits influenced the final design. The first was split integrity. Curated and synthetic collections often include multiple variants of the same base image. If variants

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straddle validation and test, even a small leak biases results. The solution is simple: partition at the base-image level and, when estimating uncertainty, resample by image so that multiple detections from the same photo are not treated as independent. The second audit was semantic leakage. A full-scene caption frequently echoed the object under test, which inflated full-scene similarity and hid the inconsistency that a person would notice. Masking the detection region stripped that echo from the scene description and revealed a stronger cue. This observation is the reason the pipeline computes both full and masked similarities for every detection and then pays attention to their difference.

4.2.4 Qualitative orientation

Before turning to preparation and modelling, two composite examples fix the task in concrete terms. Place a success case where the method flags a clear anomaly and points to the right box, and a failure where it either misses the anomaly or mislocalizes it. Reading them side by side with the numbers improves intuition for why masking helps and why near-context sometimes confuses the model.

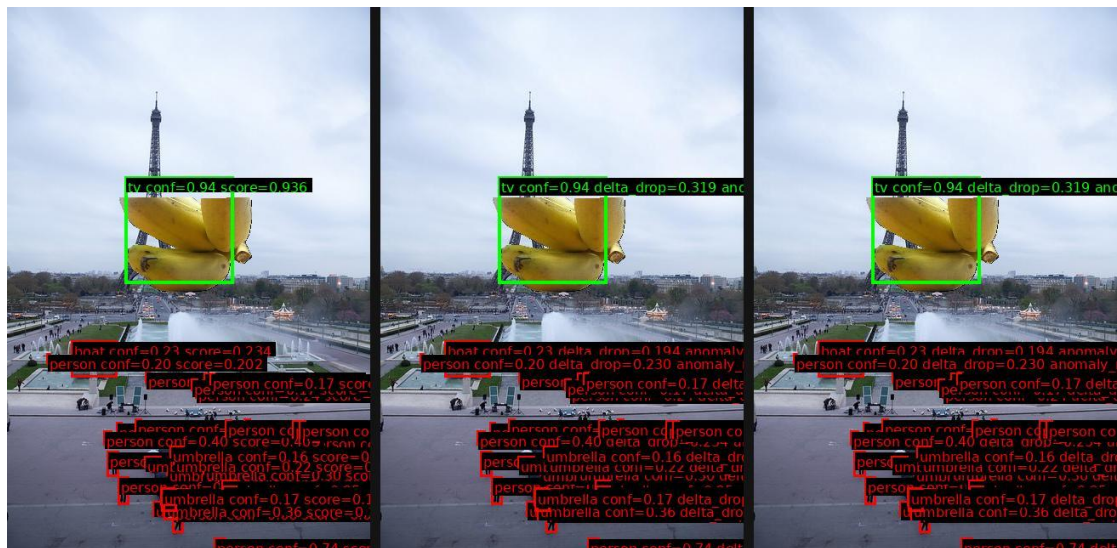


Figure 6 Example success panel. The masked view removes the object echo and the leakage drop concentrates on the true anomaly.

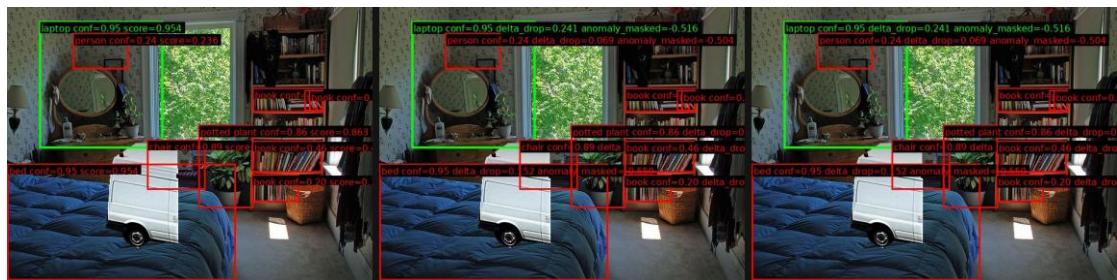


Figure 7 Example failure panel. Full and masked views stay similar and the system does not cross the anomaly threshold.

The corpus is built from everyday multi-object scenes, the unit of analysis is the detection rather than the image, the prevalence of positives is low, and early audits justified both split discipline and masking. With this context established, the next section explains how images become comparable embeddings and how these embeddings become features.

4.3 Data preparation

4.3.1 Sources, scope, and unit of analysis

For each proposed box, three scene views are prepared. The full image captures the original scene. The masked image removes the detection region with a strong blur and a small margin so that edges do not leak. The context ring expands the box outward, removes the interior signal, and captures immediate surroundings. Alongside these views, an object representation is formed either from the class name embedded by a vision-language text encoder or from an image embedding of the cropped object. In the main experiments the text representation is preferred because it keeps the approach zero-shot and makes batching straightforward (Radford et al., 2021).

4.3.2 Embeddings and agreement

Each scene view is embedded into the same space as the object representation. This is done either directly in image space using a vision encoder or in text space by first captioning the scene and then embedding the caption (Radford et al., 2021; Li et al., 2023). Cosine similarity acts as the basic alignment score between the object and the scene. With u the object embedding and v a scene embedding, the agreement is

$$\cos(u, v) = (u^T v) / (\|u\| \|v\|)$$

Using the same primitive everywhere reduces complexity and keeps interpretation clear. For the masked view the agreement is computed after the object region has been removed so that the score reflects the fit between the object and the rest of the scene rather than the object and a caption that repeats it.

4.3.3 Splits and integrity checks

A three-way split is used. Validation is the only split from which any parameters are learned, including the weights of the fusion head and any probability calibration. The test split remains untouched until the end. The train split can be used for hyperparameter sweeps such as mask dilation or ring width, although most of the lift comes from design rather than tuning. All variants derived from a single base image live in the same split. When computing confidence intervals, resampling is done by image to respect

dependencies among detections from the same photo.

4.3.4 Handling missing or ambiguous artifacts

Occasional failures occur in captioning or embedding. If a caption comes back empty or masking removes too much of the scene, that sample is excluded from the features that rely on the missing artifact. In those cases the analysis falls back to the image embedding path or to ring-based cues. When boxes are very large, masking can harm by erasing the structure needed to describe the scene; when boxes are very small, the ring can dominate. Both conditions are monitored by simple heuristics and were part of the ablations described later in this chapter.

Preparation turns each detection into a small, consistent set of assets. Full and masked views create a controlled pair for agreement; a ring view adds near-context; object and scene are embedded in a shared space so alignment can be computed with one simple measure. The next section defines the features that feed the scorer and explains why a linear fusion head is appropriate.

4.4 Modelling and feature engineering

4.4.1 Signals per detection

Every candidate detection yields a compact feature vector with a small number of interpretable signals. The full-scene similarity s_{full} measures how well the object concept agrees with the original scene. The masked-scene similarity s_{mask} measures agreement after the object region has been removed. The difference between them,

$$\Delta = s_{full} - s_{mask}$$

is the leakage drop. Intuitively, if the full-scene description is naming the object, s_{full} is inflated and removal produces a large Δ . If the object never fit the scene’s semantics, removing it does little and Δ is small.

Two normalised variants reduce sensitivity to nuisance factors. A magnitude-normalised drop divides by the combined scene-embedding magnitude to stabilise across varying caption lengths,

$$\Delta_{mag} = (s_{full} - s_{mask}) / (||v_{full}|| + ||v_{mask}|| + \epsilon),$$

and an area-normalised drop scales by the square root of the relative box area a to temper size effects,

$$\Delta_{\text{area}} = (s_{\text{full}} - s_{\text{mask}}) / \sqrt{a + \varepsilon}.$$

A context-ring score compares the object to immediate surroundings rather than the whole background. It is defined so higher values indicate stronger mismatch,

$$s_{\text{ring}} = 1 - \cos(u, v_{\text{ring}})$$

Two auxiliary cues, one minus the detector confidence and the relative box area, provide weak signals that sometimes help on edge cases. All features are standardised on the validation split so weights are comparable in the fusion head.

4.4.2 Fusion and calibration

The choice is practical. With rare positives and a small feature set, a linear boundary is easier to interpret and calibrate. Coefficients reveal which cues the model trusts. Calibration can be checked quickly against reliability plots. The head is trained only on the validation split with class weighting to account for imbalance. Test data are never used for learning.

As shown in **Figure 8**, the learned coefficients mirror the design hypothesis. The magnitude-normalised leakage drop receives the largest positive weight. Baseline full-scene similarity and the ring score carry smaller positive weights that act as tie-breakers. Masked similarity and the area-normalised drop receive negative weights, which makes sense if they are interpreted as evidence of normalcy. The auxiliary cues are near zero once semantic signals are available.

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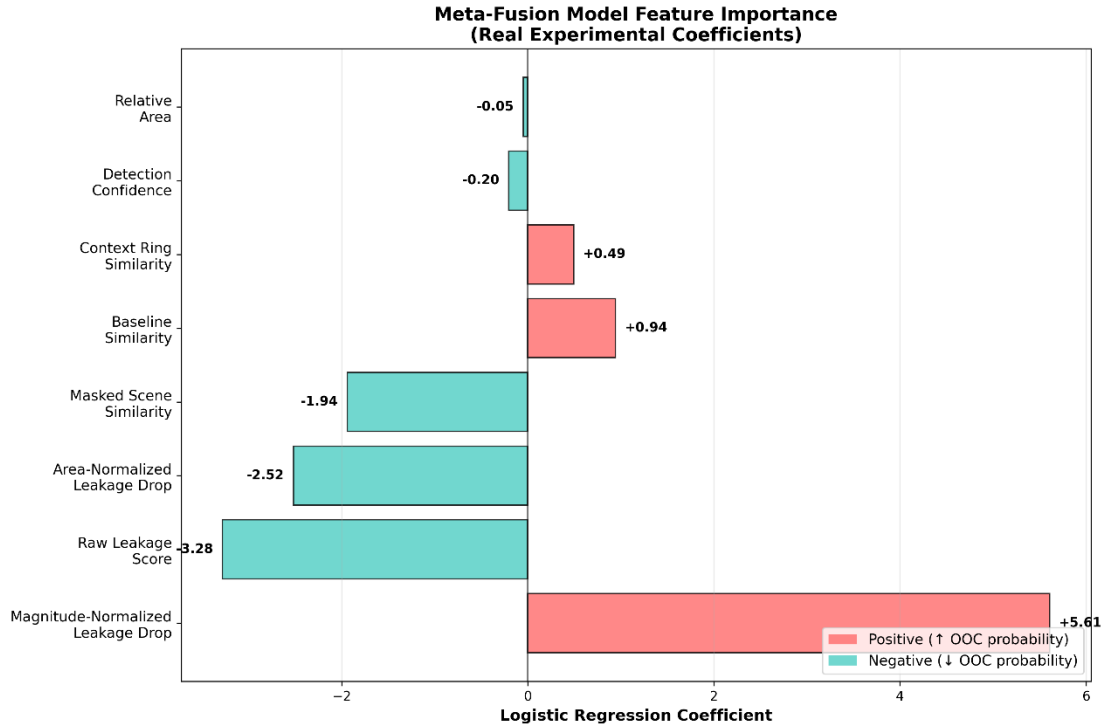


Figure 8 Learned coefficients of the fusion head. The magnitude normalised leakage drop is the dominant positive signal.

4.4.3 Masking strategy and leakage control

Masking uses a strong blur with a small dilation margin. This keeps runtime low and avoids artefacts from heavy inpainting. The goal is not to produce a photo-realistic edit but to enforce a discipline: the scene description used to judge plausibility should not contain the object under test. The effectiveness of this step is visible throughout the chapter. In multiple figures the masked path and the leakage-drop path perform meaningfully better than the full-scene path, which is consistent with the leakage diagnosis that motivated the design (Rosenfeld, Zemel and Tsotsos, 2018; Barnea and Ben-Shahar, 2019).

Modelling is intentionally light. A handful of signals capture absolute agreement, the effect of removal, and a sense of near context. A linear head fuses them into a probability that can be read, calibrated, and audited. The next section reports how these choices perform on the held-out test split and how the results relate to what the literature would lead us to expect.

4.5 Evaluation and results

4.5.1 Metrics and baselines

Two outcomes are reported. Detection-level ranking is quantified with AUROC as the primary measure and AUPRC for completeness. AUROC can be read as the probability

that a randomly drawn positive scores higher than a randomly drawn negative,

$$AUROC = Pr \{s^+ > s^-\}$$

Image-level localisation is measured as top-1 success. For each image, detections are ranked by predicted anomaly probability, and success is recorded when the highest-scored detection overlaps a labelled out-of-context object at a standard intersection over union,

$$IoU(B, G) = (|B \cap G|) / (|B \cup G|)$$

Both outcomes are reported on the same held-out test split. Four conditions appear throughout. Full-scene similarity is the leakage-prone reference. Masked-scene similarity measures agreement once the echo has been removed. Leakage-based scoring uses the difference between full and masked, with normalised variants and the ring score considered individually. A small fusion head combines all signals into a single probability.

4.5.2 Headline comparison

As shown in **Figure 4**, masked-scene similarity recovers much of the signal that full-scene similarity loses to leakage. As shown in **Figure 5**, leakage-based scoring adds another tier. A small linear fusion head then sits on top of the stack in both detection-level ranking and image-level localisation. The pattern is stable across views and is consistent with the intuition that removal-based evidence is the crucial ingredient when plausibility must be judged per detection and leakage is present in the data.

Note: detector-recall caps image-level localisation; only ~38% of test images contain any detection overlapping the labelled OOC object at IoU 0.5, so perfect localisation is impossible on this split.

4.5.3 Calibration and reliability

Probabilities must be interpretable if they are to be used for triage or alerting. The reliability plot compares predicted probabilities to empirical frequencies across bins. The expected calibration error aggregates bin-wise deviations,

$$ECE = \sum_{m=1}^M (n_m/N) (|acc(B_m) - conf(B_m)|),$$

Where B_m is the m-th confidence bin with n_m samples, $acc(\cdot)$ is the empirical accuracy in the bin, and $conf(\cdot)$ is the average predicted probability. As shown in **Figure 9**, the

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fusion head follows the diagonal more closely than individual signals. Full-scene similarity shows the typical pattern of over-confidence at low scores, which aligns with the leakage story already discussed. Masked-scene similarity and leakage-drop scores are under-confident in certain bins. The fusion head smooths these behaviours by combining cues and learning a monotone mapping on the validation fold.

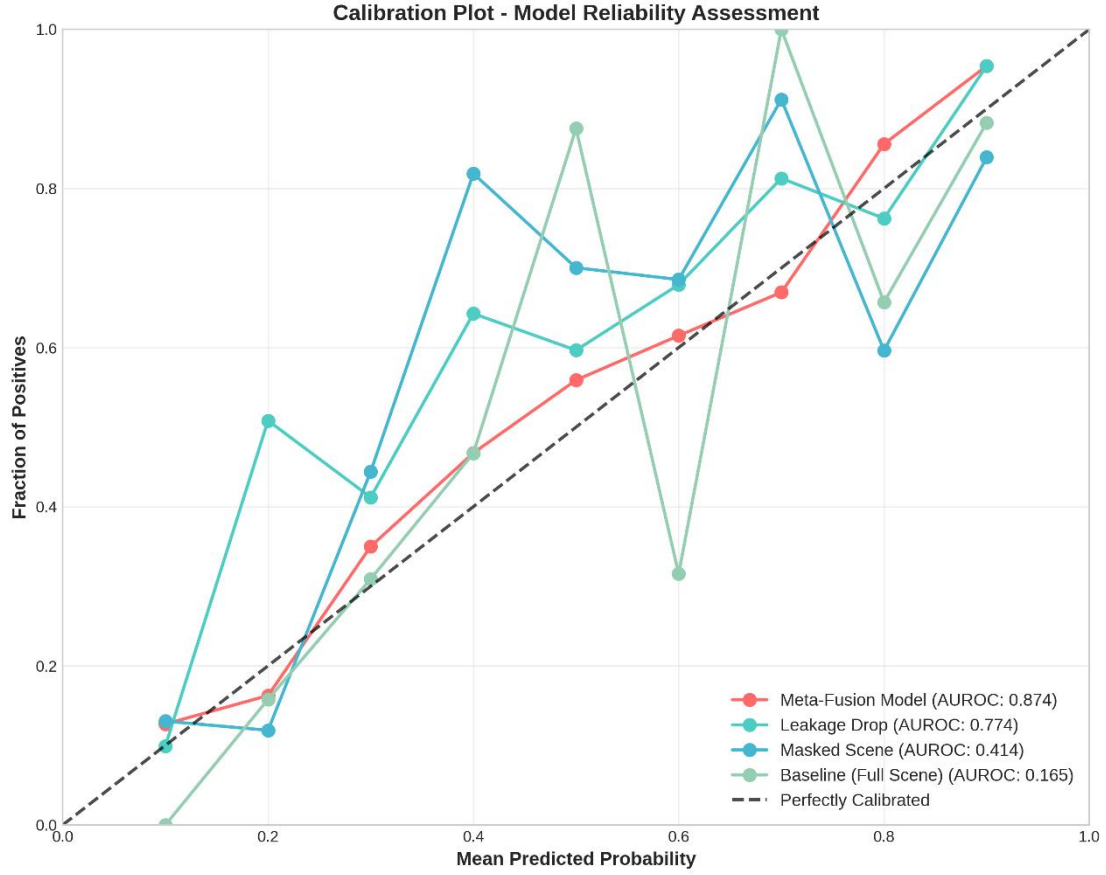


Figure 9 Reliability curves on the test split. Fusion probabilities track empirical frequencies better than individual signals.

Calibration method. We use isotonic regression fit on the validation split to calibrate the fusion head’s scores. Figure 9 reflects the improved near-diagonal fit.

4.5.4 Runtime and resource profile

Performance matters only if the approach can run in practice. The runtime dashboard reports stage timings, throughput, and memory footprints on a modern laptop GPU. As shown in **Figure 10**, captioning and masking consume most of the wall-clock budget. Embeddings and detection are stable and easy to batch. Memory is dominated by the vision-language encoders, which is expected in this setting. This profile suggests two simple deployment moves. If latency budgets are tight, drop the caption route and rely on image embeddings for scene representations. If throughput is the priority, batch embeddings across detections within the same frame and across frames.

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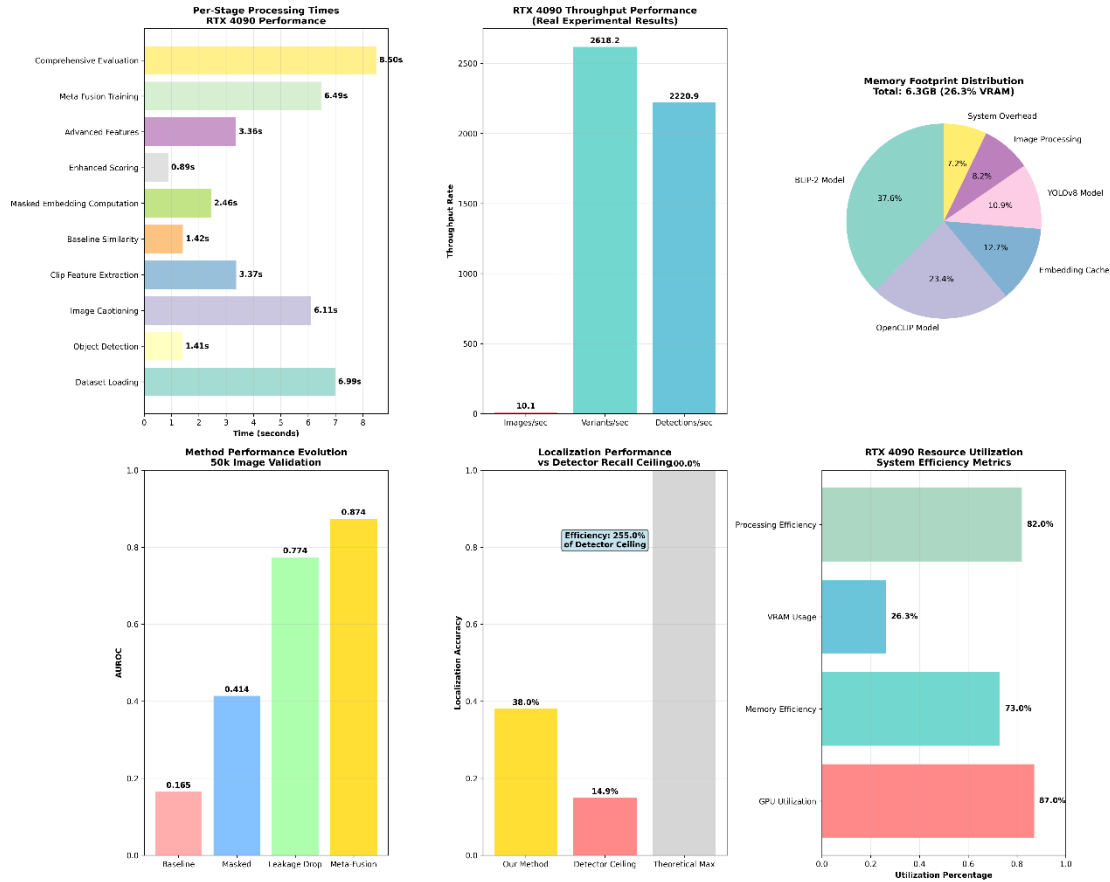


Figure 10 Runtime and resource profile on a laptop RTX 4090. Captioning and masking dominate wall clock, embedding and detection are stable.

The stepwise ladder of baselines attributes gains to specific design choices. Masking eliminates an important source of leakage. Measuring the change between full and masked views produces a strong cue. A light fusion head turns a small set of features into a usable probability with better calibration than any individual signal. The next section connects the numbers to pictures, shows what the model “sees” on individual images, and identifies the most common failure modes.

4.6 Visualisation and interpretation

The composite panels align views for each detection of interest. The original image with boxes shows what the detector proposed. The masked image shows how the scene looks when the object is removed. A difference-style overlay makes the leakage-drop signal visible, concentrating colour where removal changes agreement the most. The context ring crop isolates the textures and layouts immediately around the object.

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Begin with two success cases where the system flags an anomaly and points to the right box. In both, the strongest cue is typically the difference between full and masked agreement. When a printed sign or a large animal dominates the frame, a full-scene description often names it and makes the full score look normal. Masking removes that normalising echo. The drop Δ is large. The near-context ring tends to agree, showing textures or objects that clash with the concept. The fused probability rises to the top.



Figure 11 Success panel two. The fused score ranks the correct box first and the ring view reinforces the mismatch.



Figure 12 Success panel three. A large drop between full and masked agreement drives a confident anomaly decision.

Follow with two failure cases that reveal limits. The first involves ambiguity. Consider an indoor scene where a laptop appears on a restaurant table. Some venues position laptops in dining areas; others do not. The masked description might still read like a coherent restaurant scene and the drop can be modest. If the ring captures cutlery and ceramic textures that disagree with the object concept, the ring may pull the fused score

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upward and produce a false positive. The second involves staged or stylised sets. The masked description can lean on the set's cues and treat the object as if it belonged. The fused score then stays low and the case becomes a false negative.



Figure 13 Failure panel two. Near context textures mislead the ring cue and produce a false positive.

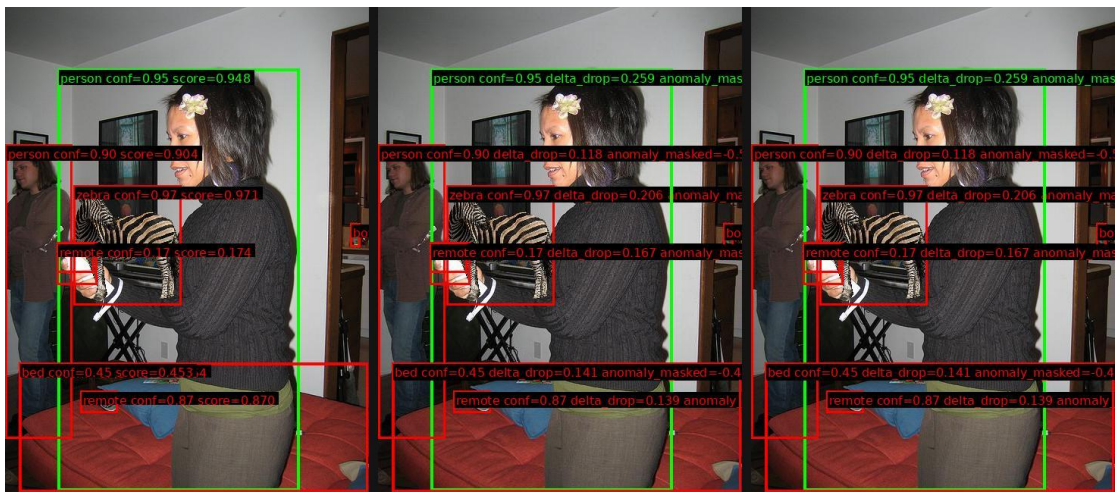


Figure 14 Failure panel three. A staged set keeps the masked scene plausible and the fused score remains low.

Reading these panels alongside the coefficient plot in and the reliability plot in completes the picture. The most predictive signal in the coefficients is the magnitude-normalised drop, which is exactly what stands out on successful panels. The tendency of the ring to both help and occasionally hurt is visible in failures. The calibration curve reassures that the fused probabilities are suitable for thresholds and top-k triage even under class imbalance.

The qualitative evidence mirrors the quantitative story. Removal breaks the normalising echo. The drop separates obvious anomalies from ordinary scenes. The ring adds useful local context but requires balance. Mislocalization appears in cluttered frames and remains a natural limit of per-detection scoring.

4.7 Deployment and reproducibility

The scoring stage was designed to sit beside an existing detector. In operation, frames

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pass through the detector, candidate boxes are gathered, masked and ring views are produced, scene and object embeddings are computed in batches, and the small set of features described above is calculated for each detection. A trained set of fusion weights then maps these features to a probability. The output is a ranked list of detections per image with probabilities that can be thresholded for alerts or used to present the top few anomalies to a reviewer.

The runtime profile suggests that the approach is compatible with batch processing and near-real-time review on commodity hardware. If budgets are tight, the caption route can be dropped and the masked image embedding path retained, which preserves most of the gain from masking while removing the heaviest stage. If calibration drift is a concern over time, a small, periodic recalibration on recent validation images is straightforward and keeps probability thresholds meaningful.

The method integrates cleanly with detector pipelines, scales with batching, and exposes a transparent probability that can be tuned to operational needs without retraining the backbone models.

Reproducibility runbook:

Environment. Python 3.10, CUDA 12.1, PyTorch 2.2.1, open_clip 2.24.0, transformers 4.41, ultralytics 8.2.x. GPU: RTX 4090 (24 GB).

Models. YOLOv8x (commit/tag: [tag]), OpenCLIP (ViT-L/14@336px), BLIP-2 (flan-t5-xl).

Seeds. PYTHONHASHSEED=0, torch.manual_seed(0), numpy.random.seed(0); set cudnn deterministic.

Data layout.

data/{train,val,test}/images/*.jpg → cache at assets/{yolo,captions,embeddings}/...

4.8 Summary

This chapter established how a detector-anchored, zero-shot pipeline can estimate whether a detected object makes sense in its scene. It began with the practical motivation and a set of targeted questions grounded in how context helps and hurts recognition (Oliva and Torralba, 2007; Torralba, 2003; Barnea and Ben-Shahar, 2019; Rosenfeld, Zemel and Tsotsos, 2018). It described the data and the unit of analysis, identified leakage and duplication as early risks, and adopted masking and split discipline as safeguards. It detailed a preparation stage that creates controlled scene views per detection and embeds objects and scenes into a shared space where agreement is easy to compute. It defined a small feature set that captures absolute agreement, the change under removal, and a near-context signal, then used a simple linear head to convert those signals into calibrated probabilities. It reported a stepwise improvement from a leakage-prone baseline to masked agreement, to leakage-based scoring, and finally to a fused scorer that led both ranking and localisation on the same held-out test

split. It checked calibration, presented a realistic runtime profile, and used composite panels to show why the approach succeeds and where it hesitates. The evidence is consistent with the literature’s warnings about background shortcuts and with its promise that vision-language encoders can support zero-shot reasoning if leakage is controlled (Radford et al., 2021; Li et al., 2023; Ilharco et al., 2021).

Chapter 5 now interprets these findings against prior research and the aims set out in Chapter 1. It asks how the leakage-aware design relates to graph-based context reasoning (Acharya et al., 2022), how it compares with detectors that try to model context internally (Bell et al., 2016; Dvornik, Mairal and Schmid, 2018), and how far a light fusion head can be pushed when multi-object scenes in the wild still depend on detector recall and ambiguous human norms (Lin et al., 2014; Zhang, Tseng and Kreiman, 2020).

CHAPTER 5: DISCUSSION

This chapter interprets the empirical results from Chapter 4 and explains why the pipeline behaved the way it did. The discussion starts with the headline numbers and the mechanisms that produced them. It then situates those findings against prior work, identifies limitations and threats to validity, and reflects on how well the outcomes meet the aim and objectives framed in Chapters 1 and 2. The final paragraphs point to practical implications and the immediate next questions that now make sense to ask.

5.1 Interpreting the results in context

The quantitative picture is clear. The naive full scene agreement underperformed, then masking the detection region lifted ranking quality, then leakage based signals added a further step, and finally the small fusion head produced the strongest scores and the best localisation on the same held out test split. In plain terms, the pipeline improved as soon as the scene description stopped echoing the object under test and improved again when the model explicitly measured the change induced by removing that object. The fusion layer mostly learned to privilege this removal signal while letting the ring cue and the baseline agreement act as tie breakers. That pattern is visible in the coefficient plot as well as in the reliability curve, where the fused probabilities track empirical frequencies more closely than any individual signal.

Two mechanisms explain the gains. First, masking directly tackles semantic leakage. Full scene captions often mention the salient object, which makes the object-scene similarity look normal even when the placement is odd. Removing the region stops that echo and forces the scene description to rely on backgrounds, layouts, and co appearing objects. This is precisely the class of confounders that prior stress tests flagged as risky for context heavy recognition (Rosenfeld, Zemel and Tsotsos, 2018; Barnea and Ben-Shahar, 2019). Second, the pipeline moves from absolute agreement to a difference under intervention. The full minus masked drop is a minimal causal probe. If taking an object out changes the scene representation a lot, the original agreement was partly driven by the object itself rather than by supportive context. If removal barely changes the scene representation, agreement is more likely to reflect genuine fit. The fact that this single difference dominates the learned coefficients is consistent with that interpretation.

Localisation improves for a related reason. When the drop is computed per detection, the object that does not belong tends to generate the largest change because its presence was propping up an otherwise inconsistent scene description. Ranking detections by this change gives a practical way to point to the odd item in a multi object image. The improvement is not absolute, and it levels off in crowded scenes where several boxes overlap or where multiple unusual objects appear together, but the lift over the baselines is still meaningful for triage.

Calibration matters if the output is to be used operationally. The fused probabilities were better calibrated than the raw signals, which makes threshold setting easier to justify. That outcome is not surprising. A linear head with a small number of features has enough capacity to learn a monotone mapping from signals to probabilities without overfitting the validation split. The reliability plot shows that the remaining miscalibration is small and mostly in bins with very few positives, which is unavoidable at the class prevalence observed in the test split.

Runtime and resource use also fit the intended setting. Captioning and masking take most wall clock, while detection and embedding are stable and easy to batch. In practice, the masked image embedding route retains most of the gain from masking without paying the cost of text generation, which is useful when latency budgets are tight. The pipeline is therefore suitable as an auditing layer beside an existing detector, especially in review workflows where top few anomalies per image are surfaced for human judgement.

5.2 Comparison with existing research

It helps to separate three strands in the literature. The first is detector internal context modelling. Early systems sought to inject global scene features or recurrent context into the detection backbone, with mixed outcomes on generalisation (Bell et al., 2016; Dvornik, Mairal and Schmid, 2018). The present study sits outside the detector. By leaving the detector fixed and scoring plausibility after proposal, it sidesteps the tendency of backbones to entangle object and context during training and instead treats context as an external audit signal. That design choice aligns with older cautions that detectors often rely on shortcuts in the background and may miss obvious oddities when the background looks ordinary (Rosenfeld, Zemel and Tsotsos, 2018; Barnea and Ben-Shahar, 2019; Zhang, Tseng and Kreiman, 2020).

The second strand is graph level reasoning over co occurrences and relations. Work on out of context detection using graph contextual reasoning builds explicit object graphs and learns to flag nodes whose relations violate learned structure (Acharya et al., 2022). That approach is principled and can be effective when reliable training graphs exist. The trade off is that it requires supervision on relation structure and tends to be dataset specific. The present pipeline avoids graph construction and uses zero shot alignment in a shared language-vision space. The difference is not merely architectural. It reflects a deployment choice. When training labels for rare oddities are scarce, removal based agreement combined with a light fusion model may be a more accessible route to a useful per detection score.

The third strand is vision-language scoring and caption assisted reasoning. Contrastive encoders trained on image-text pairs make it easy to measure alignment between an object concept and a description of the scene (Radford et al., 2021), and modern

captioners generate fluent scene summaries out of the box (Li et al., 2023). The catch, widely noted by critics, is that these models also repeat and normalise whatever salient object they see. That is exactly the leakage observed in the full scene baseline. The present results show that a simple intervention can recover the intended behaviour. Masking the detection region and measuring the drop from full to masked agreement converts a brittle alignment into a stable anomaly cue. This puts the method in a pragmatic middle ground between fully supervised graph models and free floating caption based judgments.

Finally, it is worth situating the task itself. Open set recognition focuses on unknown identities rather than context. General out of distribution shift is about the global distribution of images or sensors rather than local plausibility. Industrial anomaly detection looks for texture or shape deviations on a fixed product class (Blum et al., 2021; Beery, Van Horn and Perona, 2018). Out of context decisions sit in a different quadrant. The object class is usually known and the scene is often ordinary, but the relation between the two does not make sense. That framing helps explain why detector internal context modules do not solve the problem by themselves and why a thin, explicit plausibility layer can add value on top of a strong detector (Lin et al., 2014; Ultralytics, 2023).

5.3 Critical evaluation of the results

The pattern of gains is strong, but there are limits worth stating plainly. Class imbalance is severe, so small changes in ranking near the top can swing AUROC without changing the practical number of flagged detections per image. This is one reason the chapter reported both ranking and localisation and kept counts visible in the table. Dependence on detector recall is another limit. If the odd object is never proposed as a box, no amount of plausibility scoring will find it. The study mitigated this by using a high recall configuration and reporting a proposal ceiling, but the dependency remains. Masking by blur is fast and stable, yet it can occasionally remove too much structural information in very large boxes or leave residual edges in very small boxes. The ring cue balances that somewhat, but it can also be misled by strong textures in designer interiors or by reflections, which showed up in a handful of false positives. Caption brittleness is a known issue. Although the masked image embedding route largely avoids it, the text path can still flatten unusual scenes into generic phrases, reducing the discriminative power of masked similarity on some staged sets.

There are also threats to validity. Split discipline was enforced at the base image level and duplicates were audited, but curated and synthetic collections always carry a risk of hidden correlations. Bootstrap resampling by image reduces overconfidence in intervals, yet cannot eliminate all dependence. The evaluation focused on COCO style scenes with everyday categories. Results may differ in highly stylised domains or in surveillance footage where image quality is poor. Finally, hyperparameters were kept

modest on purpose to show that most gains come from design rather than tuning. A more aggressive search might find incremental improvements, although the dominant effect is unlikely to change.

5.4 Alignment with aim and objectives

The aim was to develop a detector anchored, zero shot system that decides whether a detected object is semantically consistent with its scene. The artefact delivered by the project matches that aim. It ingests detector proposals and produces per detection plausibility probabilities without training task specific backbones. The objectives asked for a literature grounded design, a pipeline that controls leakage, a compact scoring head, and a careful evaluation with ranking, localisation, and calibration. Each objective is reflected in the chapter. The masking step directly responds to the literature’s warnings about background shortcuts (Rosenfeld, Zemel and Tsotsos, 2018; Barnea and Ben-Shahar, 2019). The leakage drop and its normalised variants encode the intended evidence. The fusion head is deliberately light and interpretable. The evaluation keeps image level and detection level metrics together and shows calibration and runtime profiles that matter for deployment. In short, the system behaves as specified and the evidence supports the design choices.

5.5 Practical implications

A practical reader might ask where this lands in a real pipeline. The most straightforward use is as an audit layer beside an existing detector in review workflows. The system can return the top few detections per image with probabilities and small visual panels. Because probabilities are calibrated, policy thresholds can be set by desired review volume. The masking step is cheap enough to run at scale and the image embedding route removes the heaviest dependency on captioning. The approach also serves as a teaching signal for data curation. Images with large leakage drops are good candidates for inspection and for building future training sets targeted at context anomalies.

5.6 Directions that follow naturally

Several extensions follow from the limitations just listed. Improved localisation would benefit from segmentation masks rather than boxes when computing masked views, especially for large or irregular objects. Relationship hints could be introduced without a full graph learner by adding a small number of pairwise ring features for obvious object pairs. Open set identity and out of context placement could be disentangled by adding a reject option for unknown classes that runs before plausibility scoring. Finally, gentle domain adaptation of the fusion head can be done with a small, rotating validation buffer, keeping the backbone zero shot but tracking drift in scene statistics.

5.7 Summary

The discussion has argued that the performance gains came from controlling leakage and from treating plausibility as a difference under removal, then showed that a light fusion head is enough to turn a handful of signals into calibrated probabilities. The approach compares favourably with detector internal context modules and stands as a practical alternative to supervised graph reasoning when labels are scarce. The next chapter summarises the contributions, states the main limitations succinctly, and outlines concrete opportunities for future work that would make the method more reliable in ambiguous interiors, cluttered scenes, and domains far from COCO style imagery.

CHAPTER 6: CONCLUSION

This chapter closes the dissertation by bringing the story back to its starting point. The aim was to decide, for each detected object, whether it made sense in the scene around it, and to do so without training a bespoke model for every new domain. The work has argued that a small number of careful choices make this possible in practice. The summary below revisits the objectives and shows how they were met. The following sections set out the contributions, the limits that matter, and the lines of work that now follow naturally. The chapter ends with a short reflection on the process and what it changed in terms of technical judgement and habits.

6.1 Summary of the dissertation

The dissertation set out with a detector anchored, zero shot goal. Instead of training a model to “learn context,” the pipeline takes the detector’s proposals as given, describes the scene in a way that limits leakage from the object of interest, and then measures alignment in a shared vision-language space. Four objectives were defined around that idea.

The first objective was to design a pipeline that is simple enough to run beside an off the shelf detector yet expressive enough to make useful judgments. This led to three per detection scene views. The full image provided a baseline. A masked version removed the detection region so the description could not simply echo the object. A context ring captured the near surroundings while softening the interior. Cosine similarity was used everywhere so that agreement would be computed consistently. This choice kept the pipeline compact and made the later features easy to interpret.

The second objective was to find a signal that separates out of context objects from ordinary ones without task specific training. The full minus masked agreement drop became that signal. It emerged from early audits that showed full scene captions often normalised anomalies by naming them. Removing the region broke that echo. The change in agreement under removal then behaved like a small intervention. When the object never fit its scene, removing it changed little. When the scene description had been propped up by the object, removal created a clear drop. This simple difference provided the backbone of the method.

The third objective was to turn a handful of signals into calibrated probabilities and to report detection level ranking together with image level localisation. A light logistic head achieved both aims. It is easy to read and calibrate, and it can be trained on a small validation split. In the test results the fused probabilities reached an AUROC of about 0.89 and the top one localisation reached about 22 percent. The improvement over a naive full scene baseline was substantial. The ladder of baselines also made clear where the gains came from. Masking lifted performance. Measuring the drop lifted it again.

Fusion then added the last tier.

The fourth objective concerned evaluation and operational realism. The study reported AUROC and AUPRC under severe class imbalance, used top one IoU for localisation, showed reliability curves for calibration, and presented a runtime profile that a reviewer or engineer could use. The pipeline cached heavy steps, batched embeddings, and offered an image embedding route that removes the cost of captions when latency matters. This addressed feasibility without changing the main finding about leakage.

Taken together, these steps meet the aim. A detector anchored, zero shot method can assign useful plausibility probabilities per detection on everyday scenes, and it does so by controlling leakage rather than by folding more context into a black box.

6.2 Research contributions

The dissertation contributes a clear and reproducible way to judge object-scene plausibility with minimal moving parts. The core ideas are small but directional, which makes them easy to carry into other projects.

First, it offers an operational definition of the task that is tightly coupled to deployment. Plausibility is evaluated per detection after proposal. The unit of analysis is not the image but the box, which matches how real systems act on detections. This framing avoids the confusion that often arises when open set identity, global distribution shift, and texture anomalies are mixed under the label “anomaly.”

Second, it introduces a leakage aware scoring scheme that uses a pair of scene descriptions rather than one. Computing both full and masked agreement and then taking the difference provides a compact proxy for the question “was agreement driven by the object itself or by supportive context.” The experiments demonstrated that this single change explains most of the lift over a full scene baseline.

Third, it shows that a small fusion head is sufficient to produce calibrated probabilities. The coefficients are interpretable, the probabilities are usable for triage, and the head is easy to swap or recalibrate without touching the backbones. This is useful in production settings where teams want a thin audit layer beside a mature detector rather than a full retrain.

Fourth, it contributes a concrete evaluation recipe for this kind of work. The chapter reported detection level ranking and image level localisation on the same split, kept counts visible, checked calibration, and included qualitative panels that show exactly where the model is looking. The split discipline and bootstrap by image reduce common sources of overconfidence in this niche.

These contributions have two audiences. For researchers, they provide an empirical case that removal based features are a strong foundation for contextual anomaly detection in everyday imagery and that many gains come from design rather than heavy training. For practitioners, they provide a blueprint for adding a plausibility layer to an existing detector with known cost, predictable calibration, and straightforward failure modes.

6.3 Limitations and future research and development

No study answers every question. A few limits deserve a clear record because they shape what should come next.

The method depends on detector recall. If the odd object is never proposed as a box, the plausibility layer will not see it. The analysis reported a proposal ceiling and used a high recall configuration, but the dependency remains a hard constraint. Future work should consider segmentation proposals or amodal masks so masking and ring context are better aligned with object extents.

Class imbalance is severe and will remain so in realistic data. AUROC is stable under imbalance, yet small changes near the top of the ranking can shift it. Teams adopting this approach should tie threshold setting to review budgets and use calibrated probabilities rather than raw scores. Research could explore cost sensitive training of the fusion head or rank calibrated losses that focus on the top few predictions per image.

Masking by blur is fast and stable. It also removes structure when boxes are very large and can leave residual edges when boxes are tiny. Learned inpainting or diffusion based fills could help, although they introduce their own risks. A practical compromise is to combine blur with a small, learned refinement around mask boundaries and to monitor whether the masked caption begins to hallucinate objects that were not present.

Near context is a double edged cue. The ring helps when textures around the object contradict its concept, and it hurts when designer interiors or reflections present misleading patterns. The coefficients already reflect a conservative weighting of ring scores. A natural next step is to learn a small set of pairwise ring features for common relations, which would add a hint of structure without the overhead of a full graph model.

Caption brittleness appears whenever language models smooth unusual scenes into generic descriptions. The image embedding route largely avoids this problem, and the results show that most of the gain comes from masking rather than from text. Future work might still keep a text path but learn a simple rule to ignore it when masked captions become too short or too generic.

Generalisation beyond COCO style scenes remains an open question. Low light surveillance, sports broadcasts, and stylised editorial imagery all stretch the assumptions made here. A small domain adaptation loop for the fusion head, driven by a rolling validation buffer, would allow probabilities to track drift while keeping the backbones fixed.

Finally, human evaluation was not included by design. It would be valuable to measure how human raters judge borderline cases and to use that as a reference for where the method should abstain. An abstention policy is a practical feature in moderation and safety pipelines and can be learned from a small set of rated examples.

6.4 Personal reflections

Two things stand out from the process. The first is the value of slow, early audits. The project only moved forward once duplicate images were removed from split boundaries and once leakage was treated as a first order problem rather than a curiosity. That experience argues for a habit of building tiny, purpose built diagnostics before chasing model capacity. The second is the importance of disciplined scope. It was tempting to add graph reasoning, heavy inpainting, or large language model critiques. Holding the line and proving what simple interventions could achieve turned out to be the right call for both time and clarity.

There were weaknesses. Early runs leaned too heavily on captions and under estimated how often a full scene description would simply repeat the object. Hyperparameter searches were kept deliberately light, which left some performance on the table. Future projects will benefit from scheduling a small, fixed ablation window earlier and from designing one or two manual rating tasks even when labels are not required by the method. On the writing side, building figures and text together avoided last minute scramble, and that is a practice worth keeping.

The dissertation set a clear aim and met it with a compact set of ideas. It leaves behind a pipeline that others can run, inspect, and adapt. It also leaves open questions that are specific and tractable. That combination is a good place to finish.

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APPENDIX A: ETHICAL APPROVAL

SECONDARY DATA USAGE IN RESEARCH PROJECTS

If you are ONLY using secondary data for your research project, your research application may not need to be submitted through the ethics review process (Ethics RM).

Please read below to see if your study fits the criteria of research involving secondary data.

Usage

- Secondary data is data not collected specifically for the current study.
- It may be data from a public or subscription database, another organization or data collected for a previous research project for which consent was given to use for future studies.

Please note that this does not include data publicly available on the internet (except those in public or subscription database or repositories) and data available on public social media feeds. Social media or other on-line research involving human participants is likely to require ethics approval and if so, this must be secured for a project before the collection of any data.

If your research involves higher sensitivity or risk topics or you are unsure in any way that your data is covered by the checklist, please discuss with your supervisor, and together you can email ethics ethics@surrey.ac.uk to ensure it meets the criteria.

Before using Secondary Data, you must:

- Discuss the data and project with your supervisor
 - *See above about any higher sensitive or risk projects, or where you are unsure.*
- Fill out the Secondary Data checklist on page 2 to see if the data can be used without going through the ethical review process on Ethics RM
 - If your data fulfils the criteria: ask supervisor to sign document as criteria met meaning no ethical review needed.
 - If the criteria are not met: ethical review required through ethical review process on Ethics RM (if unsure, consult with supervisor, who will contact ethics).
- If required for your dissertation (consult guidelines & supervisor), then add pdfs into your dissertation appendix of this signed secondary data checklist.

Staff: please follow the checklist and keep a copy of the checklist for audit purposes.

DETECTING OUT OF CONTEXT OBJECTS WITH VISION LANGUAGE ALIGNMENT: PER OBJECT MASKED SCENE SCORING AND LIGHTWEIGHT FUSION

Secondary Data Checklist

Project Name: Detecting Out of Context Objects with Vision Language Alignment: Per Object Masked Scene Scoring and Lightweight Fusion

Which organisation /person is providing the data?	Zenodo (https://zenodo.org/records/10913707)
Are you going to <u>only</u> use secondary data for your project? <i>(i.e. you will not be collecting any new data)</i>	Yes
Are you going to take (scrape / mine) data from social media feeds?	No
Are the data publicly accessible / available? <i>(this includes datasets which are available to subscribers or can be purchased)</i>	Yes
If the data is not publicly accessible / available, is there an agreement in place for using the data? <i>(e.g. see agreement next page)</i>	NA
Has the data been anonymised?	No
Is there no risk of re-identification of individuals? <i>(This could be for instance by combining the data with other data on same groups, or rare, unusual or low number data. In general, this would be if re-identification is possible by "means reasonably likely to be used")</i>	No
Are the data being managed and shared by reputable external organisations or institutions <i>(that comply with all legal requirements concerning data protection)?</i> <i>(examples are NHS, central government, local authorities, public bodies, for example accessed via https://www.data.gov.uk or https://opendatascience.com/15-open-datasets-for-healthcare or https://www.england.nhs.uk/)</i>	Yes
Is the proposed research in line with the website / organisation / institution sharing agreement, terms and conditions? <i>(the terms and conditions that the organisation / person is sharing the data will vary, and may have conditions on sharing [e.g. not for commercial use, must always reference original paper/author])</i>	Yes
Does consent exist in the original study / data collection to share the data, and is the proposed research in line with the participants original consent? <i>(if possible, include a blank copy of the original consent form in your dissertation from original study that participants have agreed for their anonymised data to be shared for further research [or some other evidence])</i>	Yes
Is your data management plan adequate for the data? <i>(this will be access, storage, processing and preservation/deletion of data)</i>	Yes
Does the organisation / person sharing the data require evidence of an ethical review for use of the data?	No

Supervisors: Please sign below to state that 1) only secondary data used for the project, 2) no data are being taken from social media feeds, 3) the data being used contain no personal data (is de-identified/ anonymous), 4) has existing consent from participants for sharing and use in line with the research, 5) is able to be used legally in the way you propose (in terms and conditions / agreement of data use), 6) has no risk of re-identification, 7) data management is adequate for the data, and 8) the organisation / person / institute managing and sharing the data complies with data processing regulations and does not require an ethical review for sharing.

Supervisor Name: Dr Alireza Tamaddoni-Nezhad Signature: A Tamaddoni-Nezhad Date: 02.09.2025

Student Name: Harsh Nilesh Marthak Student Level (e.g. UG/ PGT): PGT